COMPUTER SCIENCE | GAZI UNIVERSITY

C++ Lab Report

Lab 3

Variables, data types, type casting and arithmetic operators

NAME	**Areesha Farooq**
ROLL NO	**11**
SECTION	**D**
PROGRAM	**Computer Science**
UNIVERSITY	**Gazi University**

Contents

1. Arithmetic Expressions and Operator Precedence — Page 2
2. Reading Age, Pay and Section — Page 3
3. Weight from Units and Mass — Page 4
4. Sum of Two Numbers — Page 5
5. Sum and Average of Five Numbers — Page 6
6. Division with Type Casting — Page 7
7. Float Assigned to an Integer — Page 8
8. Units Each with static_cast — Page 9
9. Math Tutor with Random Numbers — Page 10

1 Arithmetic Expressions and Operator Precedence

Program 1 of 9

CODE

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << 2 + 3 << endl;
    cout << 5 * 4 + 2 << endl;
    cout << 4 * 5 << endl;
    cout << 12 / 2 << endl;
    cout << 6 * 7 + 4 << endl;
    cout << 5 * 6 << endl;
    cout << 10 % 5 << endl;
    cout << 9 / 3 << endl;
    cout << 4 * 4 << endl;

    return 0;
}
```

OUTPUT

```
5
22
20
6
46
30
0
3
16

[Program finished]
```

NOTE

Nine cout statements evaluate expressions with +, *, / and %. Multiplication and division run before addition, so 5 * 4 + 2 gives 22 and 6 * 7 + 4 gives 46. The % operator returns the remainder, so 10 % 5 is 0.

2 Reading Age, Pay and Section

Program 2 of 9

CODE

```cpp
#include <iostream>
using namespace std;

int main()
{
    int age;
    double pay;
    char section;

    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section;

    cout << "Age = " << age << endl;
    cout << "Pay = " << pay << endl;
    cout << "Section = " << section << endl;

    return 0;
}
```

OUTPUT



NOTE

Declares an int, a double and a char, then reads all three with a single cin statement and prints them back. The output screenshot shows the program waiting at the input prompt.

3 Weight from Units and Mass

Program 3 of 9

CODE

```cpp
#include <iostream>
using namespace std;

int main()
{
    int units;
    float mass;
    double weight;

    cout << "Enter units: ";
    cin >> units;

    cout << "Enter mass: ";
    cin >> mass;

    weight = mass * units;

    cout << "Weight = " << weight << endl;

    return 0;
}
```

OUTPUT



NOTE

Reads units (int) and mass (float), then stores mass * units in a double called weight and prints it. The output screenshot shows the program waiting at the first input prompt.

4 Sum of Two Numbers

Program 4 of 9

CODE

```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is
" << sum << endl;

    return 0;
}
```

OUTPUT



NOTE

Takes two numbers as double values, adds them into sum and displays the result. The output screenshot shows the program waiting for the first number.

5 Sum and Average of Five Numbers

Program 5 of 9

CODE

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1 = 28;
    int num2 = 32;
    int num3 = 37;
    int num4 = 24;
    int num5 = 33;

    int sum;
    float average;

    sum = num1 + num2 + num3 + num4 + num5;

    average = sum / 5.0;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

OUTPUT



NOTE

Adds five integers (28, 32, 37, 24, 33) to get 154. Dividing by 5.0 instead of 5 keeps the decimal part, so the average is 30.8.

6 Division with Type Casting

Program 6 of 9

CODE

```cpp

#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;

    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";

    cin >> number1 >> number2;

    quotient = static_cast<float>(number1) / number2;

    cout << "Quotient = " << quotient << endl;

    return 0;
}
```

OUTPUT



NOTE

Reads two integers and divides them. static_cast<float>(number1) converts the first number to float before dividing, so the quotient keeps its decimal part instead of being cut off like integer division.

7 Float Assigned to an Integer

Program 7 of 9

CODE

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b = 2;
    float c = 4.2;

    a = b * c;

    cout << "Value of a = " << a << endl;

    return 0;
}
```

OUTPUT

```
Value of a = 8

[Program finished]
```

NOTE

b * c is 2 * 4.2 = 8.4, but a is an int, so the decimal part is dropped when it is stored. The program prints Value of a = 8.

8 Units Each with static_cast

Program 8 of 9

CODE

```cpp
#include <iostream>
using namespace std;

int main()
{
    int qty = 10;
    int salesReps = 3;
    float unitsEach;

    unitsEach = static_cast<float>(qty) / salesReps;

    cout << "Units each = " << unitsEach << endl;

    return 0;
}
```

OUTPUT



NOTE

Divides 10 items between 3 sales reps. Casting qty to float first gives 3.33333, where plain integer division would give 3.

9 Math Tutor with Random Numbers

Program 9 of 9

CODE

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    srand(time(0));

    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;
    int answer;

    cout << "Math Tutor" << endl;
    cout << "Solve the following problem:" << endl;
    cout << endl;

    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "---------" << endl;

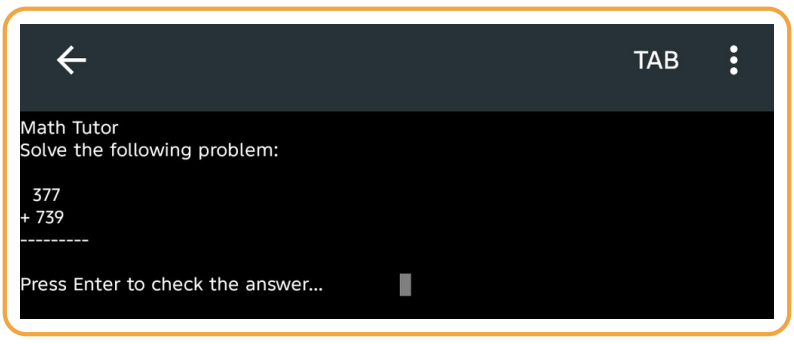
    cout << "\nPress Enter to check the answer...";
    cin.get();

    answer = num1 + num2;

    cout << "\nCorrect Answer:" << endl;
    cout << "  " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "---------" << endl;
    cout << "  " << answer << endl;

    return 0;
}
```

OUTPUT



NOTE

srand(time(0)) seeds the generator and rand() % 900 + 100 makes two random 3-digit numbers. The program shows an addition problem, waits for Enter with cin.get(), then prints the correct answer.